

Morgan Bleasdale

mjb125@ic.ac.uk

www.mbleasdale.com

+44 7512 205702

Education

Imperial College London

September 2025 – Present

MEng Electrical and Electronic Engineering

On track for First-Class Honours (graduation 2029)

Kings Monkton School, Cardiff

A Levels: Maths (A), Further Maths (A*), Physics (A*), Computer Science (A), EPQ (A*)* September 2023 – July 2025

*GCSEs: 11 A*s, 1 A and 1 Distinction (Including Maths and English)* September 2021 – July 2023

Awards: Principal's Prize for Academic Achievement, KS5 Academic Achievement Award, Jean Morris Prize for Excellence in Mathematics, Pupil Leadership Prize, McCarthy Prize for Business

Experience

Vishay Intertechnology, Inc

Newport, UK

Semiconductor Engineering Summer Internship

July 2026 – Present

- Diagnosed incorrect Active Area Deposition control limit assignment across 190 MOSFET parts, by identifying critical parameters; recalculated and reassigned limits to create a data-driven recipe-selection framework.
- Implemented upper specification limits to end-of-line BVDSS testing to prevent out-of-spec MOSFETs being shipped undetected, using 8D methodology; presented to Process Change Board for integration in facility.
- Investigated a silicon staining defect affecting ~3% of production wafers by analysing focused ion beam (FIB) failure analysis images and comparing to fab tool data to identify potential root causes.
- Updated documentation describing silicon carbide (SiC) diode manufacturing process.

UKESF Semiconductor Talent Award

London, UK

Talent Award Recipient and Mentee

September 2025 – Present

- Nationally recognised for academic excellence and future potential in semiconductor engineering, awarded a competitive, merit-based scholarship (£11,000 in first 2 years, with continued funding in subsequent years).
- Attending mentorship sessions from UK semiconductor figures, including the IET and Silicon Catalyst Europe.

National Aeronautics and Space Administration (NASA)

Cape Canaveral, Florida, USA

VP of Engineering, International Space Settlement Design Competition

July 2024 – August 2025

- Selected twice to represent the UK (top 12 of 1000+ applicants), leading team of 60+ engineering students to design a space station. **2024: Winner** (Lunar data centre), **2025: Runner-up** (Venusian Mining Settlement).
- Led overall technical design, managing 4 engineering departments with the systems engineering V-Model and a requirements traceability matrix to ensure 50-slide proposal was cohesive and complied with RFP.
- Directed Q&A section of presentation in front of panel of NASA and SpaceX engineers, and 250+ spectators.

Engineering Education Scheme Wales

Cardiff, UK

Project Coordinator and Award Winner

October 2023 – December 2024

- Designed a sustainable housing estate for 500+ residents, with a grid-tied hybrid power microgrid, intelligent EV charging infrastructure, and decentralised geothermal heating system with ground source heat pumps.
- Led team of 6 using an iterative Agile approach to present 73-page technical report to industry professionals.
- Awarded Best Use of STEM for Sustainability by Welsh Government and named runner-up as **Welsh Engineering Student of the Year 2024**; achieved Gold CREST award for engineering project excellence.

Recent Projects

RISC CPU in SystemVerilog with custom ISA

June 2026 – July 2026

16-bit Harvard-architecture RISC CPU in SystemVerilog, following the design principles of ARM architecture.

- Implemented 8 ALU operations and a flag system (NZCV) that supports jumps and carry-chained operations.
- Flag-based conditional jumping (16 conditions, derived from NZCV status flags set by ALU operations), enabling structured control flow such as loops, conditionals, and subroutine calls/returns.
- Built load/store architecture with dedicated memory instructions, between RAM and 8x16-bit register file.
- Verified design with 100% pass-rate in SystemVerilog testbenches, and an example multi-word 16x16-bit software multiplication program to demonstrate functionality.

Multi-sensor remote-controlled Wi-Fi Lunar Rover

May 2026 – June 2026

Built an RC rover that surveys a lunar environment, detecting RF, IR, ultrasonic signals, and magnetic fields.

- Built radio subsystem with a tuned LC coil antenna, inverting amplifier, envelope detector, and hysteresis comparator to demodulate an 89 kHz ASK-modulated carrier into binary UART-framed ASCII data.
- Implemented differential-drive rover movement via PWM speed control and direction; integrated with an HTTP server/webpage over Wi-Fi for remote driving and live telemetry.
- Led project management (6 people) using Agile sprints and authored the 62-page technical report.

Discrete BJT Operational Amplifier (LTSpice)

February 2026 – March 2026

Designed an operational amplifier from individual BJTs, following a differential-gain-power stage architecture.

- Designed a differential amplifier (long-tailed pair) with a Wilson current mirror active load for high differential gain, low common mode gain, and a high output impedance for improved CMRR.
- Implemented a common-emitter amplifier stage with current mirror active load to boost voltage gain.
- Developed class AB push-pull power amplifier to increase output current and reduce crossover distortion.
- Diagnosed high-frequency instability via AC analysis of phase response (negative phase margin); stabilised with a compensation capacitor to bypass common-emitter stage at high frequencies.

Skills

Digital Design & Architecture

Verilog/SystemVerilog, FPGA (Intel Quartus), CPU Architecture, CMOS Transistor-Level Design

Analogue & Embedded Systems

Analogue Circuit Design (Amplifiers and Filters), PCB Design, RF Systems, Arduino, Raspberry Pi

Software

C++, C#, Python, JavaScript, TypeScript, React (Next.js, Vue.js), SQL, Supabase, Vercel, OAuth

Volunteering

UK Space Design Competition

October 2025 – Present

- Provided technical advice and pastoral support to groups of 40-60 student competitors (Years 10-13) as they worked in groups to create and present engineering proposals over 1 or 2-day events.

Kings Monkton School, Head Student

September 2024 – July 2025

- Raised £1,000+ through charity events and initiatives, including managing partnerships with organisations.
- Organised school-wide community and cultural events, aimed at engaging students with real-world problems beyond academic study, such as a climate debate (70+ participants).